

CASCATA PORTAL

v2.0 Requirements, Gap Analysis, Test Specification & Roadmap

UPDATED WITH PM CLARIFICATIONS (22 March 2026)

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Sources: PM Video Walkthrough (40 min), Granola Meeting Notes, Gemini Transcript Export,
PM Roadmap, PM Slack Clarifications (22 Mar 2026)

Purpose: Cursor IDE implementation context + PM playback + acceptance test checklist

1. Executive Summary

This document captures **every requirement** from the PM's 40-minute video walkthrough and subsequent roadmap discussion, cross-referenced against the current Cascata Portal codebase (GitHub: ecmistry/cascata-portal, v1.1.0). It incorporates **all PM clarifications received via Slack on 22 March 2026** which resolved all 8 open questions and resulted in significant changes to the RAG system design, averaging methodology, and data requirements.

The document serves three purposes: (1) a playback document for PM validation, (2) a Cursor IDE implementation spec with exact file references, and (3) a comprehensive acceptance test checklist with 48 test cases to verify all requirements are met.

1.1 Key PM Clarifications (22 March 2026)

The following PM decisions materially changed the implementation plan:

- **Single RAG system only (Phase 1):** RAG the model's cascade outputs against actuals. Do NOT RAG the R-score yet. R-score is displayed as a number without colour coding.
- **RAG targets = model output:** Targets are derived directly from the cascade model's predictions. No separate targets/quota table needed. The model's forecast for a quarter IS the target; actuals are measured against it.
- **R-score granularity:** 6-quarter rolling average computed per region and globally. NOT per pod/motion (insufficient data points).
- **Data requirement:** 18 months (6 quarters) of HubSpot data required. Can reduce to 12 months (4 quarters) minimum. No need to handle edge cases for fewer than 4 quarters.
- **Region naming:** Keep NORAM/EMESA. The AI transcript mistranslated NORAM as 'North'. No display name changes needed.
- **Historical cutoff confirmed:** Current quarter = forecast. Everything prior = actualised. Not configurable.
- **Customer identification:** Asked during onboarding setup wizard, same pattern as other configuration questions.

Key finding: The core three-stage cascade model (SQL → OCR → OWR → Revenue) is already implemented. All Contact and Deal HubSpot fields are already fetched. The main gaps are: (1) hierarchical expandable drill-down UI, (2) Pearson R-score engine (display only, no RAG), (3) performance RAG system (model output vs actuals), (4) 6-quarter one-time average for future projections, (5) dynamic pipeline cover ratio and ACV calculations, and (6) Company object integration for Phases 2–4.

2. Current Codebase Architecture

The Cascata Portal is a production-deployed web application hosted on AWS EC2 at cascata.online. This section provides the full technical context needed for implementation.

2.1 Tech Stack

Frontend: React 19, TypeScript, Vite 7, Tailwind CSS 4, tRPC 11, TanStack React Query, Recharts, ReactFlow, Wouter, shadcn/ui.

Backend: Node.js 20+, Express 4, tRPC 11, Drizzle ORM, MariaDB/MySQL 10.5+.

Integrations: HubSpot CRM API v3 (private app token), Google BigQuery (project reporting-299920, optional).

Infrastructure: AWS EC2, Nginx reverse proxy, PM2 process manager, Let's Encrypt SSL, pnpm 10+.

2.2 Backend Files

`server/cascadeEngine.ts` (354 lines) — Core forecast engine. SQL→Opportunity conversion with time distribution, Opportunity→Win timing with probability matrices, revenue computation. Uses per-quarter actuals-derived conversion rates when available, falling back to overall averages. Primary file for 6-quarter one-time average, dynamic pipeline cover ratio, and dynamic ACV enhancements.

`server/cascadeSheet.ts` (324 lines) — Server-side two-panel cascade calculator mirroring Excel. Left panel: SQL cascade (timing probabilities × SQL counts × conversion rate → cascaded opportunities). Right panel: Opp cascade (win timing probabilities × total opps → won deals). Needs actual values alongside model values.

`server/hubspotSync.ts` (868 lines) — HubSpot ELT sync engine. Contacts and deals via CRM API v3, mapped to cascade model structures. Delta sync via `lastmodifieddate`. Currently fetches Contacts and Deals ONLY. Company object fetch entirely missing — must be added for Phases 2–4.

`server/routers.ts` (29K) — All tRPC API routes: auth, company, forecast, region, sqlType, sqlHistory, conversionRate, dealEconomics, timeDistribution, actuals, scenarios, dashboard, cascadeSheet. Needs new procedures for hierarchical data, R-scores, and RAG status.

`server/db.ts` (31K) — Database access layer. All CRUD operations via Drizzle ORM.

`server/hubspot-client.ts` (5K) — HubSpot API client for property discovery and data fetching (Configure Cascata page).

2.3 Frontend Files

`client/src/pages/Dashboard.tsx` (730 lines) — Main dashboard with KPI cards, filter bar (company/region/sqlType/year/quarter), revenue charts, regional breakdown, conversion funnel. Needs most significant overhaul for hierarchical drill-down interface.

`client/src/pages/CascadeSheet.tsx` (539 lines) — Cascade matrix with SQL timing probability diagonals, cascaded opportunity values, OWR cascade. Supports `/cascade/:motion/:region` routing. Needs Model|Actual dual-column enhancement.

`client/src/pages/Setup.tsx` (29K) — Multi-step onboarding wizard. Needs new step for Company object customer identification field (PM confirmed: “ask up front like we do for other questions”).

`client/src/pages/WhatIfAnalysis.tsx` (19K) and `Scenarios.tsx` (11K) — What-if and scenario management. Foundation for Phase 4 scenario planning.

`client/src/pages/Performance.tsx` (12K) — Actual vs predicted comparison. Foundation for R-score display.

`client/src/pages/DataQuality.tsx` (49K) — Data quality reporting after HubSpot sync.

`client/src/components/DashboardLayout.tsx` (24K) — Layout wrapper with sidebar navigation, header, auth checks.

`client/src/App.tsx` (175 lines) — Routing configuration using Wouter.

2.4 Database Schema

`drizzle/schema.ts` defines:

- **users** — accounts with bcrypt password hashing, roles (admin/user), session tracking
- **companies** — organisation models with BigQuery config and syncConfig JSON (HubSpot property mappings)
- **regions** — geographic pods (NORAM, EMESA North, EMESA South, etc.) per company
- **sqlTypes** — lead source types (Inbound, Outbound, ILO, Event, Partner) per company
- **sqlHistory** — historical SQL volumes by region/type/year/quarter
- **conversionRates** — `oppCoverageRatio` + `winRateNew` + `winRateUpsell` in basis points, per region/type
- **dealEconomics** — average contract values (`acvNew` + `acvUpsell`) in cents, per region
- **timeDistributions** — SQL→Opp timing (`sameQ/nextQ/twoQ %`) + Opp→Deal win timing JSON
- **forecasts** — cached forecast results (`predictedSqls`, `predictedOpps`, `predictedRevenue`) — NOTE: these serve as RAG targets per PM clarification
- **actuals** — actual performance (`actualSqls`, `actualOpps`, `actualRevenue`) — NOTE: missing `actualWins`
- **scenarios** — saved what-if scenarios with adjustment parameters and cached results
- **dataQualityReports** — sync quality metrics (coverage %, skipped records, unmapped values)

2.5 Constants (`shared/const.ts`)

Default cascade parameters: same-quarter conversion 89% (8900bp), next-quarter 10% (1000bp), two-quarter 1% (100bp). Default coverage ratio 5% (500bp). Default win rate 25% (2500bp). Default ACV \$100,000 (10000000 cents). Revenue split: 70% new business / 30% upsell. Forecast window: 5 years from Q1 2024.

3. PM Implementation Roadmap

The PM has defined a **strict sequential implementation order** where each phase builds the foundation for the next. The video walkthrough requirements (Phase 1: new bookings) must be completed first.

3.1 Four-Phase Build Order

Phase	Scope	Status	Priority	Depends On
Phase 1	New Bookings: complete cascade model (SQL→OCR→OWR→revenue), hierarchical drill-down UI, Model vs Actual dual tracking, Pearson R-score (display only), performance RAG system (model output vs actuals), 6-quarter one-time average for future projections	PARTIAL	P1 - HIGH	None
Phase 2	Upsells: upsell-specific OCR and OWR cascades (separate from new business), upsell attach rate × customer base × avg upsell ACV = upsell bookings. Requires Company object integration.	NEW	P1 - HIGH	Phase 1 complete
Phase 3	Churn: customer retention/loss rate tracking per quarter/region. Churn revenue impact = churned customers × avg ACV. Customer identification via Company object.	NEW	P2 - MEDIUM	Phase 2 complete
Phase 4	Net Bookings & ARR: Net = New Bookings + Upsells - Churn. Rolling ARR projection. Scenario planning (model 20% QoQ/YoY SQL growth). Compare baseline/optimistic/conservative.	NEW	P2 - MEDIUM	Phase 3 complete

3.2 RAG System Framework (PM Clarified)

IMPORTANT PM CLARIFICATION: Only ONE RAG system for Phase 1. RAG the model's cascade outputs. Do NOT RAG the R-score.

Performance RAG (Phase 1): Compares the cascade model's predictions against actuals. The model's forecast output for a quarter IS the target. Actuals are measured against it. Green = actual is 90%+ of model prediction. Amber = 70–90%. Red = below 70%. Applies to SQL, OCR, OWR metrics at all hierarchy levels.

RAG target source: Targets are derived directly from the existing `forecasts` table (the model's own predictions). No separate targets or quota table is needed. This was explicitly confirmed by the PM: "RAG targets are derived directly from the cascade output."

R-score RAG (DEFERRED): The Pearson R-score will be calculated and displayed as a numeric value, but should NOT have RAG colour coding in Phase 1. PM: "Don't worry about RAG'ing R yet — just focus on RAG'ing outputs from the model."

3.3 Forecasting Methodology (PM Clarified)

Historic is always actual. All past quarters display the real data from HubSpot. No model predictions are shown for completed quarters.

Future quarters use a one-time 6-quarter average (NOT a rolling average). At forecast time, the system takes the last 6 completed quarters of actual data for each metric (SQL→OCR conversion rate, OCR→OWR win rate, pipeline cover ratio, ACV, timing distributions), computes a single arithmetic average, and applies that fixed number to all future quarter projections. This average is recalculated only when a new sync/forecast is triggered — it does not continuously roll forward.

PM clarification (verbatim): *"For SQL:OCR conv — you put all ACTUAL data in. You then look at the last 6 quarters and average it (same is done for OCR:OWR). Then this average is used for future quarters."*

Current quarter = forecast. Everything prior = actualised. This is not configurable.

3.4 Data Requirements (PM Clarified)

- **Required:** 18 months (6 quarters) of historical data in HubSpot for the model to function properly
- **Minimum fallback:** 12 months (4 quarters) if 18 months unavailable
- **No edge case handling needed:** PM confirmed there will always be sufficient historical data in HubSpot. "I suspect there won't be fewer than 6 quarters. You don't need to do this looking backward."
- At forecast time, the system averages the last 6 completed quarters of actuals once and uses that fixed average for all future projections

3.5 R-Score Granularity (PM Clarified)

R-score computed at two levels only:

- **Per region** (e.g., NORAM, EMESA North, EMESA South)
- **Globally** (weighted average across all regions)

NOT per pod/motion. PM: "I would not do it per pod, but instead do it per region and then globally." Per-motion would have insufficient data points with only 6 quarters.

3.6 Post-Implementation Capabilities

Once all four phases are complete, the portal provides:

- **Scenario planning:** Model SQL increases (20% QoQ or YoY growth), test various growth assumptions, compare baseline vs optimistic vs conservative
- **Comprehensive dashboard:** Multi-level RAG visibility (global, by acquisition method, by region), interactive spreadsheet with clickable navigation arrows
- **Full metric tracking:** SQLs, OCR, OWR, new bookings, upsells, upsell-specific OCR/OWR, churn, net bookings, ARR projections
- **Future tutorials:** PM will provide additional video walkthroughs for Phases 2 (upsells) and 3 (churn)

4. HubSpot Data Requirements

Complete audit of every HubSpot field required, cross-referenced against what the codebase already fetches.

4.1 Contact Object — All Fields Already Fetched

hubspotSync.ts line 351 fetches all required Contact properties. **No changes needed.**

Object	Property	Purpose	Status	Code Reference
Contact	admin__first_became_a_sql_date	SQL date (Stage 1 start)	EXISTS	hubspotSync.ts L49
Contact	admin__first_became_an_opportunity_date	Opp date (Stage 1 end)	EXISTS	hubspotSync.ts L52
Contact	contact_pod	Region/pod assignment	EXISTS	hubspotSync.ts L50
Contact	type_of_sql	SQL type / lead method	EXISTS	hubspotSync.ts L51
Contact	lifecyclestage	Lifecycle stage filter	EXISTS	hubspotSync.ts L351
Contact	hs_lead_status	Lead status filter	EXISTS	hubspotSync.ts L351
Contact	createdate	Fallback date / sort	EXISTS	hubspotSync.ts L351
Contact	lastmodifieddate	Delta sync filter	EXISTS	hubspotSync.ts L351

4.2 Deal Object — All Fields Already Fetched

hubspotSync.ts line 369 fetches all required Deal properties. **No changes needed.**

Object	Property	Purpose	Status	Code Reference
Deal	dealstage	Pipeline stage (won/lost filter)	EXISTS	hubspotSync.ts L369
Deal	dealtypes	New business vs upsell	EXISTS	hubspotSync.ts L369
Deal	amount	Deal value for ACV calc	EXISTS	hubspotSync.ts L55
Deal	closedate	Close date (OWR timing)	EXISTS	hubspotSync.ts L56
Deal	createdate	Create date (OWR timing)	EXISTS	hubspotSync.ts L57
Deal	deal_pod	Deal region/pod	EXISTS	hubspotSync.ts L53

Deal	type_of_sql_associated_to_deal	SQL type linked to deal	EXISTS	hubspotSync.ts L54
Deal	lastmodifieddate	Delta sync filter	EXISTS	hubspotSync.ts L369

4.3 Company Object — THE MAIN DATA GAP

CRITICAL: The current codebase has **ZERO Company object fetching**. hubspotSync.ts only fetches Contacts and Deals. The PM explicitly stated three HubSpot objects are required: Contact, Deal, AND Company.

From Granola notes (Rory): "Company: [Customer status field - needs identification from user]" and "Customer count tracking via company object status field"

From Gemini transcript (Rory): "Users must specify which HubSpot field identifies a company as a customer" and "The model depends on three primary HubSpot objects: Contact (Person), Deal, and Company"

PM clarification on customer field: "This should be a question we ask up front — like we do for other questions." Confirmed: ask during onboarding setup wizard.

Object	Property	Purpose	Status	Code Reference
Company	[user-configured field]	Identifies firm as customer (asked during onboarding)	NEW	SyncConfig + hubspotSync.ts
Company	hs_object_id	Company ID for joining to deals	NEW	hubspotSync.ts
Company	name	Company name for portal display	NEW	hubspotSync.ts
Company	createdate	When company created / became customer	NEW	hubspotSync.ts
Company	lifecyclestage	Likely candidate for customer ID field	NEW	hubspotSync.ts
Company	num_associated_deals	Deal count for activity tracking	NEW	hubspotSync.ts
Company	total_revenue	Revenue for churn impact calculation	NEW	hubspotSync.ts

4.4 Exact Code Changes for Company Object

A) SyncConfig interface (drizzle/schema.ts lines 50–73): Add `companyCustomerField: string` and `companyCustomerValues: string[]`.

B) hubspotSync.ts (after deal fetch, ~line 385): Add `fetchAllRecords('companies', companyProps, [...])` following existing Contact/Deal pattern. Process results to count customers per quarter/region.

C) Setup.tsx wizard: Add new onboarding step for Company field selection (same pattern as Contact/Deal mapping).

D) HubSpot Private App scopes: Add `crm.objects.companies.read` and `crm.schemas.companies.read`.

4.5 Phase 1 Changes (No New HubSpot Fields)

Phase 1 does NOT require any new HubSpot properties. Changes are about processing existing data:

- **Pipeline cover ratio:** Count `totalClosedWon/totalClosedLost` per Q/region/motion from existing `dealstage`. Dynamic instead of static `oppCoverageRatio`.
- **Per-quarter ACV:** `mean(amount)` for closed-won deals per Q/region/motion from existing `amount` field. Replace static `dealEconomics.acvNew`.
- **actualWins column:** Add to `actuals` table. Count of closed-won deals per Q/region/sqlType.
- **quarterlyMetrics table:** Per-quarter aggregated ratios, ACV, win/loss counts for one-time average calculation at forecast time.

5. Gap Analysis: PM Requirements vs Current Code

Each requirement classified as EXISTS, PARTIAL, or NEW. Updated to reflect PM clarifications.

5.1 Phase 1 — Cascade Engine

ID	Requirement	Status	File(s)	Priority
CM-01	Three-stage cascade: SQL → OCR → OWR → Revenue	EXISTS	cascadeEngine.ts	N/A
CM-02	SQL→OCR timestamp: admin_first_became_sql → admin_first_opportunity	EXISTS	hubspotSync.ts	N/A
CM-03	OCR→OWR filtered to closed_won + closed_lost only	EXISTS	hubspotSync.ts	N/A
CM-04	OWR timing distribution (create_date → close_date delta)	EXISTS	cascadeSheet.ts	N/A
CM-05	Dynamic pipeline cover ratio per quarter (not static config)	PARTIAL	cascadeEngine.ts	P1
CM-06	Dynamic ACV from closed-won deals per Q/region/motion	PARTIAL	hubspotSync.ts, db.ts	P1
CM-07	6-quarter one-time average for future projections (not rolling — computed once at forecast time)	NEW	cascadeEngine.ts	P1
CM-08	Historical Qs = actuals, current Q = forecast (PM confirmed, not configurable)	PARTIAL	cascadeEngine.ts	P1
CM-09	Per-quarter conversion rates from actuals when available	EXISTS	cascadeEngine.ts L171	N/A
CM-10	Only two manual inputs (SQL volume + conversion rates)	PARTIAL	cascadeEngine.ts, Setup.tsx	P2

5.2 Phase 1 — Hierarchical UI & Dashboard

ID	Requirement	Status	File(s)	Priority
UI-01	Level 1 (Global): aggregated SQL/OCR/OWR by quarter	PARTIAL	Dashboard.tsx	P1
UI-02	Level 2 (Region): expandable breakdown by regional pod	PARTIAL	Dashboard.tsx	P1
UI-03	Level 3 (Motion): granular view by lead source method	PARTIAL	CascadeSheet.tsx	P1
UI-04	Interactive expand/collapse arrows (Global→Region→Motion)	NEW	New component	P1

UI-05	All child levels sum/average to parent automatically	PARTIAL	cascadeEngine.ts	P1
UI-06	Model Actual dual columns per metric per row	NEW	New component	P1
UI-07	Color coding: purple/lavender for SQL/OCR, green for OWR	NEW	CSS + component	P2
UI-08	Interactive spreadsheet with clickable navigation	NEW	New component	P1
UI-09	Multi-level performance RAG visibility (global, method, regional)	NEW	New component	P1

5.3 Phase 1 — Validation, R-Score & RAG (Updated per PM)

ID	Requirement	Status	File(s)	Priority
VAL-01	Pearson R-score: 6-quarter rolling average (PM: per region + globally, NOT per pod)	NEW	New pearsonEngine.ts	P1
VAL-02	R-score per region and globally (PM confirmed: not per pod/motion)	NEW	pearsonEngine.ts	P1
VAL-03	Overall dashboard R-score as headline KPI (numeric display, no RAG colour)	NEW	Dashboard.tsx	P1
VAL-04	R-score displayed as number only — NO RAG colouring in Phase 1 (PM: deferred)	NEW	Dashboard.tsx	P2
VAL-05	Performance RAG: model output = target, actuals vs model. Green 90%+, Amber 70-90%, Red <70%	NEW	New ragEngine.ts	P1
VAL-06	Performance RAG at all hierarchy levels (Global, Region, Motion)	NEW	HierarchicalCascade	P1
VAL-07	Model vs Actual at OCR level	PARTIAL	actuals has actualOpps	P1
VAL-08	Model vs Actual at OWR level	NEW	Need actualWins column	P1
VAL-09	Diagnostic: identify which funnel stage diverges	NEW	New component	P2

5.4 Phase 1 — Financial Pipeline

ID	Requirement	Status	File(s)	Priority
FIN-01	pipeline_cover_ratio = (won + lost) / won	PARTIAL	cascadeEngine.ts	P1

FIN-02	$\text{anticipated_wins} = \text{OWR} / \text{pipeline_cover_ratio}$	PARTIAL	cascadeEngine.ts	P1
FIN-03	$\text{new_business_bookings} = \text{wins} \times \text{avg_ACV}$	EXISTS	cascadeEngine.ts L214	N/A
FIN-04	ACV varies by quarter (dynamic, not static)	PARTIAL	hubspotSync.ts	P1

5.5 Phase 2 — Upsells

PM will provide separate video walkthrough.

ID	Requirement	Status	File(s)	Priority
UPS-01	Upsell-specific OCR cascade (separate from new business)	NEW	cascadeEngine.ts	P1
UPS-02	Upsell-specific OWR cascade (separate from new business)	NEW	cascadeEngine.ts	P1
UPS-03	$\text{Upsell bookings} = \text{attach rate} \times \text{customer base} \times \text{avg upsell ACV}$	NEW	New module	P1
UPS-04	Customer count from Company object	NEW	hubspotSync.ts	P1
UPS-05	Dashboard: upsell metrics alongside new business	NEW	Dashboard.tsx	P2

5.6 Phase 3 — Churn

PM will provide separate video walkthrough.

ID	Requirement	Status	File(s)	Priority
CHR-01	Customer retention/loss rate per quarter/region	NEW	New churnEngine.ts	P2
CHR-02	Customer identification field in onboarding (PM: ask up front)	NEW	SyncConfig, Setup.tsx	P2
CHR-03	$\text{Churn revenue} = \text{churned customers} \times \text{avg ACV}$	NEW	churnEngine.ts	P2
CHR-04	Dashboard: churn metrics in hierarchy	NEW	Dashboard.tsx	P2

5.7 Phase 4 — Net Bookings & ARR

Synthesizes all previous phases.

ID	Requirement	Status	File(s)	Priority
NET-01	$\text{Net Bookings} = \text{New Bookings} + \text{Upsells} - \text{Churn}$	NEW	New module	P2
NET-02	ARR projection rolling forward	NEW	New module	P2

NET-03	Scenario: model 20% QoQ/YoY SQL growth	PARTIAL	WhatIfAnalysis.tsx	P2
NET-04	Compare baseline / optimistic / conservative	PARTIAL	Scenarios.tsx	P2
NET-05	Dashboard: full metric tracking (SQL through ARR)	NEW	Dashboard.tsx	P2

6. Implementation Plan for Cursor (Phase 1)

Each step lists exact files to create/modify. Load this section plus referenced files into Cursor.

6.1 Step 1: Pearson R-Score & RAG Engines

Create `server/pearsonEngine.ts`: Pearson R calculation accepting two number arrays. 6-quarter rolling window. Compute per region and globally (not per pod/motion, per PM). Returns R scores as numeric values only — no RAG colouring.

Create `server/ragEngine.ts`: Performance RAG calculator. Compares actuals against model's cascade output (from forecasts table). Single mode: Green $\geq 90\%$, Amber 70–90%, Red $< 70\%$. No separate targets table — the model's predictions ARE the targets.

Update `drizzle/schema.ts`: Add `actualWins` (int) to actuals table. Add `quarterlyMetrics` table. Add `rScoreHistory` table. NO targets table needed (PM confirmed RAG uses forecasts table).

Update `server/routers.ts`: Add `validation.getRScores` (returns numeric R per region + global) and `validation.getRagStatus` (compares forecasts vs actuals, returns RAG colour per metric/level).

6.2 Step 2: Enhanced Calculation Methodology

Update `server/cascadeEngine.ts`: Add 6-quarter one-time average function (NOT rolling — computed once at forecast time, applied to all future Qs). For each metric, take the last 6 completed quarters of actuals, compute arithmetic mean, use that fixed number for all future projections. Dynamic pipeline cover ratio = $(\text{won} + \text{lost}) / \text{won}$ per quarter for historicals, averaged for future. Dynamic ACV = $\text{mean}(\text{amount})$ from closed-won deals, same pattern.

Update `server/hubspotSync.ts`: During deal sync, compute per-quarter: `totalClosedWon`, `totalClosedLost`, pipeline cover ratio, mean ACV. Write to `quarterlyMetrics`. Add `actualWins` count to actuals upsert.

6.3 Step 3: Hierarchical Drill-Down UI

Create `client/src/components/HierarchicalCascade.tsx`: Three-level expandable tree. Each row: quarter label, SQL (Model|Actual), OCR (Model|Actual), OWR (Model|Actual), R-score (numeric, region/global only), performance RAG indicator. Expand/collapse arrows. Purple/lavender SQL/OCR, green OWR. Children roll up to parents.

Update `client/src/pages/Dashboard.tsx`: Replace flat KPIs with `HierarchicalCascade` as primary view. Keep filter bar. Add headline R-score as numeric value (no RAG colour). Add headline performance RAG summary.

Update `server/routers.ts`: Add `cascadeSheet.getHierarchicalData` returning pre-aggregated data at Global/Region/Motion levels with model + actual values, R-scores per region/global, and performance RAG per level.

6.4 Step 4: Model vs Actual in CascadeSheet

Update `client/src/pages/CascadeSheet.tsx`: Add actual columns alongside model values. Divergence indicators.

Update `server/cascadeSheet.ts`: Extend `CascadeSheetData` with `actualSqls`, `actualOpps`, `actualTotalOpps`, `actualWins` fields. Pull from `actuals` table.

7. Database Schema Changes

7.1 Modified Tables

actuals: Add `actualWins` (int, default 0) — closed-won deal count per Q/region/sqlType.

companies.syncConfig (JSON): Add `companyCustomerField` (string), `companyCustomerValues` (string[]), `rollingWindowQuarters` (int, default 6).

7.2 New Tables

quarterlyMetrics: `companyId`, `regionId`, `sqlTypeId`, `year`, `quarter`, `pipelineCoverRatio` (BP), `avgAcvNew` (cents), `avgAcvUpsell` (cents), `totalClosedWon`, `totalClosedLost`, `customerCount`. Per-quarter actuals enabling the 6-quarter one-time average computation at forecast time.

rScoreHistory: `companyId`, `metricType` (enum: ocr/owr/overall), `regionId` (nullable — null for global), `year`, `quarter`, `rScore` (decimal), `sampleSize`.

7.3 Tables NOT Needed (PM Clarified)

targets table: NOT REQUIRED. RAG targets are derived directly from the existing `forecasts` table. The model's cascade predictions serve as the targets against which actuals are compared. PM: "RAG targets are derived directly from the cascade output."

8. Calculation Formulas Reference

8.1 Three-Stage Cascade

Stage 1 (SQL→OCR): Opportunities = SQL_volume × conversion_rate. Time-distributed: same_quarter 89%, next_quarter 10%, two_quarters_later 1%.

Stage 2 (OCR→OWR): Win timing from close_date - create_date. Example: 14% same Q, 33% Q+1, 25% Q+2, 15% Q+3, 7% Q+4, 4% Q+5, 2% Q+6.

Stage 3 (Revenue): pipeline_cover_ratio = (won + lost) / won. wins = OWR / ratio. bookings = wins × avg_ACV.

8.2 Pearson Correlation Coefficient

$R = (n\sum xy - \sum x \sum y) / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$. 6-quarter rolling window. **Computed per region and globally only (not per pod/motion)**. Displayed as numeric value. No RAG colouring in Phase 1.

8.3 Performance RAG

RAG_status = actual / model_prediction. **Model prediction (from forecasts table) IS the target.** Green: ≥ 90%. Amber: 70–89%. Red: < 70%.

8.4 6-Quarter Average (One-Time, Not Rolling)

IMPORTANT DISTINCTION: This is NOT a continuously rolling average. At forecast time, the system takes the last 6 completed quarters of actuals, computes a single arithmetic mean, and applies that fixed value to all future quarters. The average is only recomputed on the next sync/forecast trigger.

For each metric (SQL→OCR conversion rate, OCR→OWR win rate, pipeline cover ratio, ACV): forecast_value = mean(actual values from most recent 6 completed quarters). Applied to ALL future quarters uniformly. **Historical quarters always use actuals (PM: “Historic is always actual”)**. Current quarter = forecast. Everything prior = actualised.

PM clarification: *“That is not a rolling average. It is a one time average used for future.”*

8.5 Future Phase Formulas

Upsell (Phase 2): attach_rate × customer_count × avg_upsell_ACV

Net Bookings (Phase 4): new_bookings + upsells - churn. ARR = current_ARR + net_bookings.

9. Comprehensive Test Specification

48 test cases covering every PM requirement. **Updated to reflect PM clarifications** (single RAG system, R-score numeric only, model output as RAG target, region+global R-score only).

9.1 Cascade Engine Tests (TC-CM)

TC	Test Case	Verification Steps	Expected Result	☐
CM-01	Three-stage cascade output	Trigger forecast with seeded data. Check SQL, opportunities, revenue per Q/region/motion.	Non-zero values for all three stages across all regions/types.	
CM-02	Correct HubSpot timestamps	Sync contacts. Verify admin_first_became_sql and admin_first_opportunity used.	Timing reflects SQL→Opp gap, not contact creation.	
CM-03	Only closed deals in OWR	Check deal fetch filter. Verify active pipeline excluded.	Deal count = sum(closed_won + closed_lost) only.	
CM-04	Dynamic pipeline cover ratio	After sync, verify ratio = (won+lost)/won per quarter.	Ratio varies by quarter, not single static value.	
CM-05	Dynamic ACV from actuals	After sync, verify ACV from actual deal amounts per Q/region/motion.	ACV varies by quarter/region. Not static config.	
CM-06	6-quarter one-time average	Seed 8 quarters. Trigger forecast. Verify future Qs all use same averaged value (not recalculated per Q).	All future Q projections use identical value = mean of last 6 completed Q actuals.	
CM-07	Historical = actuals, current Q = forecast	Navigate to completed quarter vs current quarter.	Past Qs show actuals. Current Q shows forecast. Not configurable.	
CM-08	18-month data requirement	Check system validates 6 quarters of data. Test with 4Q minimum.	System works with 4+ quarters. 6 quarters recommended.	

9.2 Hierarchical UI Tests (TC-UI)

TC	Test Case	Verification Steps	Expected Result	☐
UI-01	Level 1 Global renders	Navigate to dashboard. Check aggregated totals.	Global row with totals summing all regions.	
UI-02	Level 2 Region expand	Click expand on Global. Check region breakdown.	Region rows appear; sum to Global.	

UI-03	Level 3 Motion expand	Click expand on Region. Check motion breakdown.	Motion rows appear; sum to Region.	
UI-04	Roll-up accuracy	Expand all. Sum L3→L2→L1 manually.	Exact mathematical roll-up at every level.	
UI-05	Model Actual dual columns	Check every row for dual sub-columns.	Both Model and Actual shown per metric.	
UI-06	Purple SQL/OCR colour	Inspect SQL and OCR columns.	Purple/lavender styling.	
UI-07	Green OWR colour	Inspect OWR columns.	Green styling.	
UI-08	Navigation arrows	Click expand/collapse arrows.	Smooth toggle states.	

9.3 R-Score Tests (TC-VAL) — Updated per PM

TC	Test Case	Verification Steps	Expected Result	☐
VAL-01	R calculation correct	Seed known model+actual for 6 Qs. Manual R calc. Compare.	System R matches manual within 0.01.	
VAL-02	R per region (not per pod)	Check R-scores displayed. Verify per-region only.	R-score shown for each region and globally. NOT per motion/pod.	
VAL-03	Overall R headline (numeric)	Check dashboard headline KPI.	Single numeric R-score displayed. No RAG colour.	
VAL-04	R-score has NO RAG colouring	Verify R-score display has no green/amber/red.	R-score shown as plain number. No RAG badges.	
VAL-05	Rolling window updates	Add Q7 data. Verify Q1 dropped.	R recalculates using most recent 6 Qs.	
VAL-06	R-score with 4Q minimum	Test with only 4 quarters.	R-score calculated (reduced window). No crash.	

9.4 Performance RAG Tests (TC-RAG) — Updated per PM

IMPORTANT: RAG targets come from the model's cascade output (forecasts table), NOT a separate targets table.

TC	Test Case	Verification Steps	Expected Result	☐
RAG-01	Green: actual 95% of model prediction	Model predicts 100 SQLs. Actual = 95.	Green (95% >= 90%).	
RAG-02	Amber: actual 75% of model	Model predicts 100. Actual = 75.	Amber (75% in 70-90%).	

RAG-03	Red: actual 60% of model	Model predicts 100. Actual = 60.	Red (60% < 70%).	
RAG-04	RAG on SQL metric	Compare model's SQL prediction vs actual SQLs.	Correct RAG colour for SQL.	
RAG-05	RAG on OCR metric	Compare model's OCR prediction vs actual opps.	Correct RAG colour for OCR.	
RAG-06	RAG on OWR metric	Compare model's OWR prediction vs actual wins.	Correct RAG colour for OWR.	
RAG-07	RAG at Global level	Check Global row for performance RAG.	RAG shown on aggregate row.	
RAG-08	RAG at Region level	Expand. Check each region.	RAG per region (may differ from Global).	
RAG-09	RAG at Motion level	Expand further. Check each motion.	RAG per motion within each region.	
RAG-10	RAG source is forecasts table	Verify RAG targets come from forecasts, not separate table.	No targets table exists. Model predictions used directly.	

9.5 Financial Pipeline Tests (TC-FIN)

TC	Test Case	Verification Steps	Expected Result	☐
FIN-01	Cover ratio formula	Q with 10 won, 30 lost.	ratio = 40/10 = 4.0	
FIN-02	Anticipated wins	OWR=50, ratio=4.0.	wins = 50/4 = 12.5	
FIN-03	Bookings calc	wins=12.5, ACV=\$80K.	bookings = \$1,000,000	
FIN-04	ACV varies by quarter	Check ACV for different Qs.	Not identical; reflects actual deal sizes.	

9.6 HubSpot Integration Tests (TC-HS)

TC	Test Case	Verification Steps	Expected Result	☐
HS-01	Contact props fetched	Trigger sync. Check API logs.	All 8 contact properties included.	
HS-02	Deal props fetched	Trigger sync. Check API logs.	All 8 deal properties included.	
HS-03	Company fetch (Phase 2+)	Trigger sync. Verify Company query.	Company fetch with min 6 properties.	
HS-04	Only closed deals for OWR	Check deal fetch filter.	Only closed_won + closed_lost.	

HS-05	New vs upsell separation	Check deal processing.	Separate counts/ACV per deal type.	
HS-06	Region mapping	Sync with various pod values.	All map per contactRegionMap.	
HS-07	SQL type mapping	Sync with various type_of_sql.	All map per sqlTypeMap.	
HS-08	Quarterly metrics stored	After sync, check quarterlyMetrics.	Rows per Q/region/motion.	

9.7 Data Integrity & Edge Cases (TC-INT)

TC	Test Case	Verification Steps	Expected Result	☐
INT-01	Empty quarter	Q with 0 SQLs and 0 deals.	0 values. No NaN/Infinity.	
INT-02	Single region company	Company with 1 region, 1 motion.	Hierarchy renders correctly.	
INT-03	Partial data quarter	SQLs but no closed deals.	OCR normal. OWR = 0. No errors.	
INT-04	Conversion > 100%	More opps than SQLs (data issue).	Capped at 100%. Warning shown.	
INT-05	Perfect R-score	All model = actuals.	R = 1.0 (numeric, no RAG colour).	
INT-06	Zero variance actuals	All actuals identical.	Graceful handling. N/A or warning.	
INT-07	Cross-browser	Chrome, Firefox, Safari.	Consistent rendering.	

9.8 Future Phase Smoke Tests (TC-FUT)

TC	Test Case	Verification Steps	Expected Result	☐
FUT-01	Upsell cascade independent	Phase 2: change new business inputs.	Upsell unaffected.	
FUT-02	Customer count tracking	Phase 2: verify Company sync.	customerCount per Q.	
FUT-03	Churn calculation	Phase 3: verify formula.	Churn subtracted from net.	
FUT-04	Net bookings	Phase 4: verify composite.	New + Upsell - Churn = Net.	
FUT-05	20% QoQ scenario	Growth scenario. Check cascade.	Downstream metrics increase.	

10. PM Clarifications (Resolved 22 March 2026)

The PM (Rory Blundell) provided answers to all 8 open questions via Slack on 22 March 2026. All 8 are now fully resolved. Changes have been incorporated throughout the document.

Q#	Question	PM Answer & Impact	Status
Q1	Region naming	Keep NORAM/EMESA. AI transcript mistranslated NORAM as 'North'. No display name changes.	RESOLVED
Q2	R-score granularity	6-quarter rolling average per region and globally. NOT per pod/motion. Insufficient data points per motion.	RESOLVED
Q3	Rolling window edge case	18 months (6Q) required in HubSpot. Can reduce to 12 months (4Q). No edge case handling needed — sufficient data will always exist.	RESOLVED
Q4	Pipeline cover ratio	Same pattern as all metrics: actual per-quarter values for history, 6-quarter one-time average for future. "You put all ACTUAL data in. You then look at the last 6 quarters and average it. Then this average is used for future quarters." Not a rolling average.	RESOLVED
Q5	RAG targets source	"RAG targets are derived directly from the cascade output." No separate targets table. Model predictions ARE the targets.	RESOLVED
Q6	Customer ID field	"This should be a question we ask up front — like we do for other questions." Ask during onboarding wizard.	RESOLVED
Q7	Historical cutoff	"Correct — current quarter is forecast and everything prior is actualised." Not configurable.	RESOLVED
Q8	Two RAG systems	"Don't worry about RAG'ing R yet — just focus on RAG'ing outputs from the model." Single RAG system for Phase 1. R-score displayed as number only.	RESOLVED

11. Cursor IDE Context Instructions

Always include these files as context:

- `server/cascadeEngine.ts`, `server/cascadeSheet.ts`, `server/hubspotSync.ts`, `server/routers.ts`, `server/db.ts`
- `drizzle/schema.ts`, `shared/const.ts`
- `client/src/pages/Dashboard.tsx`, `client/src/pages/CascadeSheet.tsx`, `client/src/components/DashboardLayout.tsx`, `client/src/App.tsx`

New files to create (Phase 1):

- `server/pearsonEngine.ts` — R-score calculations (region + global, numeric only)
- `server/ragEngine.ts` — Performance RAG (model output vs actuals, single system)
- `client/src/components/HierarchicalCascade.tsx` — Three-level drill-down UI

Key PM decisions to remember when prompting Cursor:

- RAG targets = forecasts table (model predictions). No separate targets table.
- R-score: numeric display only. No RAG colours. Per region + global only.
- Historical quarters = actuals. Current quarter = forecast. Not configurable.
- 6-quarter ONE-TIME average (not rolling): compute mean of last 6Q actuals at forecast time, apply to all future Qs.
- Pipeline cover ratio, conversion rates, ACV all follow same pattern: actuals for history, 6Q average for future.
- Region names stay as NORAM/EMESA (no changes).

Prompt template:

"I am implementing Phase 1, Step [X] of the Cascata v2.0 spec. Key PM decisions: RAG targets come from the forecasts table (model output), R-score is numeric only (no RAG colour, per region + global), historical = actuals / current Q = forecast. The files are [list]. Changes needed: [paste from Section 6.X]. Maintain TypeScript, tRPC, Drizzle, Tailwind, shadcn/ui. Run pnpm check after."

— End of Document —